

Abdul Hannan Kanji

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SUMMARY

ML engineer with 10+ years in software engineering, including 4+ years of ML at Google. In Play Ads, builds ad format and asset selection models, with 39 production launches driving **cumulative nine-figure incremental annual revenue**. Seeking senior IC roles in ML / RL.

EXPERIENCE

Software Engineer (ML) | Google, Mountain View, CA

Feb 2022 — Present

- Proposed and led format-conditional inference for the Play Ads auction model, moving a deep recommendation model from one prediction per ad to one per valid ad format per query. Owned headroom analysis, design, model changes, and serving, onboarding auction and infra teams. Drove nine-figure incremental annual revenue at no measurable inference cost; advised another Google ad stack that has since adopted it.
- Replaced static score-based asset selection in Play Ads with a learned model that picks the best creative asset per ad impression. Framed the modeling task (training data, loss, asset score definition), built the model and its features, and partnered with the serving team on production rollout. Drove nine-figure incremental annual revenue.
- Shipped 39 production launches — drove 22 hands-on and led 17 — coordinating model rollouts, A/B experiments, and production handoffs across ranking, serving, and product teams.
- Own the production asset selection model and work on several other production models (factorized two-tower and fully connected) predicting clicks, installs, and conversion events. Handle on-call escalations and production outages.
- Formally mentor 3 engineers, including one who shipped their own launch and was promoted. Review launch designs, present talks and posters at internal Ads summits, and champion GenAI coding workflows for the team.

Applied Data Scientist | SwayAI, Lowell, MA

Jul 2021 — Feb 2022

- Delivered AWS SageMaker ML solutions for 3 enterprise clients — UNet anomaly detection (aeronautics), object detection serving (radiology), caller sentiment analysis (customer service) — productized as no-code platform components.

Research Engineer | Reich Lab, UMass Amherst, Amherst, MA

Jun 2020 — May 2021

- Owned the data validation, visualization, and [build pipeline](#) for the [COVID-19 Forecast Hub](#) — the official data source for the CDC COVID-19 Forecasting page.
- Rebuilt the submission validation pipeline, reducing validation time from ~20 minutes to <1 minute and unblocking weekly submissions from 50+ research teams.

Software Development Engineer | Intuit Inc., Mountain View, CA

Aug 2015 — Jul 2019

- Built the Android app shell, a shared framework packaging common Intuit services (auth, logging) into one library, mandated across the QuickBooks, TurboTax, and Mint mobile apps; shipped QuickBooks Android transactions features.
- Shipped on-device ML receipt validation using MLKit, eliminating manual transcription overhead and saving ~\$4,500/month in operational costs.

PROJECTS & OPEN SOURCE

Seldonian Safety Project — Graduate Researcher, advised by Prof. Philip Thomas (UMass Amherst), 2020–2021

- Extended the Seldonian framework (training ML models under high-confidence safety constraints, Science, 2019) with stratified sampling, increasing the probability of returning a safe solution by ~60% in worst-case binary classification settings. Released as a pip package ([docs](#), [GitHub](#)).
- Built safe policy improvement for RL: CMA-ES and cross-entropy-method policy search against high-confidence bounds on per-decision importance sampling (PDIS) off-policy estimates, parallelized on Ray.

Microsoft Fairlearn — Graduate Researcher with Microsoft Responsible AI team, 2020

- Implemented the [Equality of Opportunity](#) fairness constraint and made [ExponentiatedGradient](#)'s estimator copying more efficient; both merged into Microsoft's Fairlearn. Benchmarked against TensorFlow Constrained Optimization.

EDUCATION

University of Massachusetts, Amherst — M.S. Computer Science, Amherst, MA

Sep 2019 — Jun 2021

Coursework: Reinforcement Learning (Prof. Philip Thomas)

PES Institute of Technology — B.E. Computer Science, Bengaluru, India

Sep 2011 — Jun 2015

SKILLS

ML/AI: Production ranking & recommendation, large-scale training & serving, A/B experimentation, model quality optimization, reinforcement learning & off-policy evaluation, ML safety & fairness. **Stack:** Python, C++, Java; PyTorch, TensorFlow, TPUs, Scikit-learn, Pandas, NumPy, AWS SageMaker, Colab.